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#include <stdio.h>
#include <stdlib.h>
#define MAX 100
typedef struct {
  int items[MAX];
  int size;
} PriorityQueue;
void swap(int* a, int* b)
{
  int temp = *a;
  *a = *b;
  *b = temp;
}
void heapifyUp(PriorityQueue* pq, int index)
{
  if (index
      && pq->items[(index - 1) / 2] > pq->items[index]) {
      swap(&pq->items[(index - 1) / 2],
           &pq->items[index]);
      heapifyUp(pq, (index - 1) / 2);
  }
}
void enqueue(PriorityQueue* pq, int value)
{
  if (pq->size == MAX) {
      printf("Priority queue is full\n");
      return;
  }

  pq->items[pq->size++] = value;
  heapifyUp(pq, pq->size - 1);
}
int heapifyDown(PriorityQueue* pq, int index)
{
  int smallest = index;
  int left = 2 * index + 1;
  int right = 2 * index + 2;

  if (left < pq->size
      && pq->items[left] < pq->items[smallest])
      smallest = left;

  if (right < pq->size
      && pq->items[right] < pq->items[smallest])
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        smallest = right;

    if (smallest != index) {
        swap(&pq->items[index], &pq->items[smallest]);
        heapifyDown(pq, smallest);
    }
}
int dequeue(PriorityQueue* pq)
{
    if (!pq->size) {
        printf("Priority queue is empty\n");
        return -1;
    }

    int item = pq->items[0];
    pq->items[0] = pq->items[--pq->size];
    heapifyDown(pq, 0);
    return item;
}
int peek(PriorityQueue* pq)
{
    if (!pq->size) {
        printf("Priority queue is empty\n");
        return -1;
    }
    return pq->items[0];
}
int main()
{
    PriorityQueue pq = { { 0 }, 0 };
    enqueue(&pq, 3);
    enqueue(&pq, 2);
    enqueue(&pq, 15);
    enqueue(&pq, 5);
    enqueue(&pq, 4);
    enqueue(&pq, 45);


    printf("%d dequeued from queue\n", dequeue(&pq));

    printf("Top element is %d\n", peek(&pq));

    return 0;
}
```